

Suppose a random variable has probability density function

$$f(b) = Ae^{-(2b-1)^2}$$

where A is a constant and $-\infty < b < \infty$. What is the mean times the variance?

(a) $1/16$

(b) $1/8$

(c) $1/4$

(d) $1/2$

(e) 2

(f) 1

(g) 4

(h) $1/\sqrt{8}$

(i) $1/\sqrt{2}$

(j) $2\sqrt{2}$

(k) 8

(l) None of these.

gaussian pdf: $f_X(u) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{u-m}{\sigma}\right)^2}$

$$-(2b-1)^2 = -\frac{1}{2}\left(\frac{u-m}{\sigma}\right)^2$$

$$-\left(2\left(b-\frac{1}{2}\right)\right)^2 =$$

$$-\left(4\left(b-\frac{1}{2}\right)^2\right) =$$

$$-4\left(b-\frac{1}{2}\right)^2 = -\frac{1}{2\sigma^2}\left(u-m\right)^2 \quad u = \frac{1}{2}$$

$$-4 = -\frac{1}{2\sigma^2}$$

$$2\sigma^2 = \frac{1}{4}$$

$$\sigma^2 = \frac{1}{8}$$

$$\frac{1}{2} \cdot \frac{1}{8} = \frac{1}{16}$$

Suppose a fair 100-sided die is rolled once and the resulting value is called X . That is, $X \in \{1, 2, 3, \dots, 100\}$. What is the variance of $\sin(\pi X/2)$?

(a) $1/2$

(b) $1/4$

(c) 1

(d) 2

(e) 100

(f) 50

(g) 25

(h) $\sqrt{2}$

(i) $1/8$

(j) 4

(k) $25/2$

(l) None of these

$$\text{let } Y = \sin\left(\frac{\pi X}{2}\right)$$

$$Y: 1 \rightarrow 0 \rightarrow -1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow -1 \rightarrow 0$$

$$\text{so } P(Y=1) = \frac{1}{4}$$

$$P(Y=-1) = \frac{1}{4}$$

$$P(Y=0) = \frac{1}{2}$$

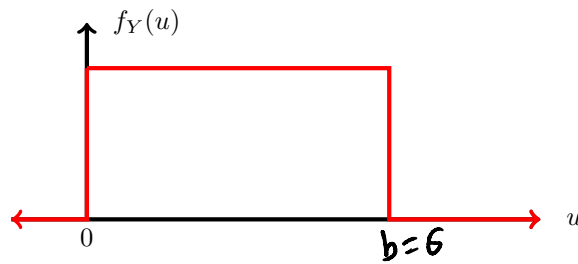
$$E[Y] = \frac{1}{4} - \frac{1}{4} = 0$$

$$E[Y^2] = \frac{1}{2}$$

$$\text{var}(Y) = \frac{1}{2} - 0^2$$

$$= \frac{1}{2}$$

Suppose Y is a random variable with variance equal to three and whose probability density function is shown below in red. What is the expected value of $e^{Y/3}$?



$$\text{Var}(Y) = 3$$

$$Y \sim U(0, b)$$

(a) $(e-1)(e+1)/2$

(b) $(e^2+1)/2$

(c) e^2+1

(d) $(e^2+1)/3$

(e) $(e^2+1)/6$

(f) $(e^2-1)/3$

(g) $(e+1)/3$

(h) $(e+1)/6$

(i) $(e-1)/3$

(j) $(e-1)/2$

(k) 3

(l) None of these.

$$f_Y(u) = \begin{cases} \frac{1}{b} & , 0 < u < b \\ 0 & , \text{else} \end{cases} \rightarrow \frac{1}{6}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

$$3 = \frac{b^2}{12}$$

$$b = 6$$

$$E[g(Y)] = \int_{-\infty}^{\infty} g(u) f_Y(u) du$$

$$[e^{Y/3}] = \int_0^6 e^{u/3} \left(\frac{1}{6}\right) du$$

$$= \frac{1}{6} \left[3e^{u/3} \right]_0^6$$

$$= \frac{1}{2} [e^2 - 1]$$

$$= \frac{1}{2} [(e-1)(e+1)]$$